



A new approach to gas flow calibration

Paul D. Levine*

Zero K Designs, San Francisco, CA 94134, United States

ARTICLE INFO

Article history:

Received 4 February 2010

Received in revised form 18 July 2010

Accepted 3 September 2010

Available online 15 September 2010

Keywords:

Gas flow calibration

Flow meter

Reference standard

ABSTRACT

Primary gas flow standards create calculable flow rates using two different techniques. Displacement systems rely upon the accurate determination of the first order change in position of an object being displaced by gas flowing at a constant pressure. Pressure, volume, temperature and time (PVTt) systems provide a nearly constant mass flow of gas to an accumulation tank of known volume over a fixed time interval. Measurements of pressure, temperature and time are used to calculate flow rate.

The Precision Gas Flow Calibrator (PGFC) described below represents a new and simplified approach to gas flow calibration. By setting the parameters which drive the exponential decay of pressure from a fixed volume a range of flow rates is generated which is linear to first order within measurement times sufficient for obtaining high precision data. Indeed, previous work has proven this method as effective as displacement based standards for an orifice flow calibration system [1].

© 2010 Elsevier Ltd. All rights reserved.

1. Introduction

Gas flow standards which use gas flowing at constant pressure to produce mechanical displacement (e.g. Piston and bell provers) require high accuracy dimensional measurements as well as high tolerance mechanical assemblies to achieve primary standard levels of precision. Through active control of gas pressure and displacement, higher order effects are minimized to enhance measurement precision. These systems are mechanical in nature and require substantial sophistication in their manufacture and utilization to achieve primary standards levels of uncertainty. They address flow ranges from $6.20 \times 10^{-7} \text{ m}^3/\text{s}$ to $2.40 \times 10^{-2} \text{ m}^3/\text{s}$ [2].

PVTt systems utilize Critical Flow Venturis to stabilize gas flow and create steady state conditions such that gas expands into a known collection volume at a nearly constant mass flow rate for a measured time interval. The gas density can only be accurately determined once thermal and pressure equilibrium have been reached. As gas

temperature varies substantially during the collection time, collection tanks need to be immersed in temperature controlled water baths to encourage a rapid approach to equilibrium. Precision measurements of volume, temperature and absolute pressure are required. These systems are typically large and complex. They address flow rates from $1.74 \times 10^{-5} \text{ m}^3/\text{s}$ to well over $100 \text{ m}^3/\text{s}$ [3].

The levels of precision obtained have firmly established piston and bell provers and PVTt systems as the recognized standards for gas flow calibration. They are used for the most part in primary standards laboratories.

The PGFC system creates calculable flows by allowing gas to flow from a pressurized fixed volume to vacuum through a high impedance leak thus creating an exponential decay of pressure in the fixed volume. The rate of pressure drop for times short compared to the exponential time constant is linear to first order at calculable levels of precision (see below). The volumetric flow rate is directly proportional to the rate of pressure drop. Thus accurate measurements of the fixed volume and the rate of pressure drop can yield high precision results. The PGFC can generate a wide range of volumetric flow rates through setting the parameters which define the exponential time constant to keep it long with respect to a specified

* Tel.: +1 415 533 8894.

E-mail address: info@zerokdesigns.com

measurement interval. This method takes full advantage of the resolution and sensitivity offered by state of the art Capacitance Diaphragm Gauge (CDG). The CDG is ideally suited to the measurement process as it is linear in response by design and is an accepted standard instrument for pressure measurement as used in national standards laboratories. As such, it should be possible to achieve uncertainty levels approaching those of primary references. There are no measurement critical moving parts, no dimensional measurements are necessary once system parameters have been established and only passive thermal control is necessary as the leak down process is, in principle, isothermal.

Pressure drop is measured differentially and is therefore virtually immune to instrument offsets. A *Symmetric Conductance Manifold (SCM)* comprised of conduits of equal length for both gas flow and pressure sensing ensures that pressure drops are uniform and thus do not affect the determination of rate of change. The entire symmetric array of gas vessels, CDGs and the SCM are mounted in a form fit polystyrene foam insulating enclosure to minimize the effects of temperature fluctuations in the environment. The flexibility and simplicity of the design suggests it can be readily implemented to provide suitable flow rates for gas flow meter calibration. The results cited in Ref. [1] obtained with an identical design concept optimized for an orifice flow vacuum standard, were in excellent agreement with those expected from pressure drop measurements and orifice conductance calculations.

The PGFC is a robust design and has the potential for providing a cost effective, high precision industrial reference for the calibration of flow measuring devices. This paper describes a PGFC designed using readily available hardware and instrumentation to address gas flow rates in the range of 1.74×10^{-8} – 1.74×10^{-5} m³/s. This is similar to the flow rates generated by small to medium piston provers and to the minimum flow addressed by PVTt systems. A realistic calculation yields a basic minimum uncertainty ($k = 1$) of 0.35%.

2. The $v\Delta p$ flowmeter principle

The flow rate Q of a constant temperature gas at pressure P from a fixed volume V through a fixed conductance C to a continuously pumped volume is:

$$Q = V \frac{dP}{dt} = -CP \quad (1)$$

Therefore, the pressure decays exponentially as:

$$P = P_0 e^{-\frac{C}{V}t} \quad (2)$$

where P_0 is the initial pressure.

The exponential function can be approximated by a Taylor series expansion in time with the coefficients in powers of C/V :

$$P = P_0 \{1 - (C/V)t + ((C/V)t)^2/2\} \quad (3)$$

It can be shown that the linear term in the expansion dominates for small changes in supply volume pressure. For example, for a pressure change of 0.5% ($P/P_0 = 0.995$)

in a fixed time interval t , the contribution from the first non-linear term in the expansion is 0.25%. An adjustable leak valve varies the conductance, and a symmetric array of cylinders allows for multiple values of supply volume. The measurement time interval can be chosen at will and is driven primarily to minimize statistical uncertainty. Setting a total pressure drop limit of 0.5% the relationship between configurable parameters becomes apparent:

$$(C/V)t \leq .005 \quad (4)$$

This basic relationship defines the criteria for addressing specific flow ranges.

3. Calibrator design and method

The PGFC consists of a symmetric array of gas cylinders connected through a SCM that includes interconnecting tubes and ports for gas inlet, outlet and pumping, and CDGs for absolute and differential pressure measurement. Valves on the ends of each cylinder allow for its use in gas supply or as pressure reference. CDGs are also connected to the manifold in a symmetric fashion to monitor the pressure of the gas in the volumes as well as the differential pressure between them.

The ability to set the leak valve conductance as well as the size of the supply volume and the supply pressure allows a time interval to be chosen which ensures the overall pressure drop does not exceed 0.5% of a given supply pressure. By constraining the data to correspond to 0.5% of the total drop, the decay itself is linear to within ~0.25%. The CDGs used in Ref. [1] exhibited typical measurement uncertainties of 0.25%. Therefore a total pressure drop of 0.5% could be considered as the practical upper limit of precision using CDG response as the measurement parameter. Values other than 0.5% can be used with corresponding changes to the deviation from linearity and thus how faithfully the decay can be represented by CDG data.

Constraining the overall drop in supply pressure to 0.5% of reading greatly reduces the sensitivity of the absolute CDGs to the rate of change. Utilizing differential CDGs whose full scale range corresponds closely to 0.5% of the absolute pressure value maximizes CDG sensitivity to yield high resolution rate data. There are many realizations possible utilizing this concept allowing for many flow rate ranges to be addressed.

The concept of the SCM ensures that all paths from gas supply to the delivery port are of equal conductance. In addition, pressure sensing lines are of equal conductance and measure pressure at the delivery port. The symmetry in conductance ensures that under steady state flow conditions, the rate of pressure decay will be the same throughout the system. The SCM is designed with high enough conductance to produce negligible pressure drops for the flow rates generated (see the uncertainty analysis for a sample calculation of pressure drop).

A specific design to address flow rates to 1.74×10^{-5} m³/s (~1000 sccm) within the constraints described above is shown schematically in Fig. 1. It is based on a symmetric array of eight 0.0038 m³ (1 gal) stainless steel gas cylinders and four CDGs connected through an SCM.

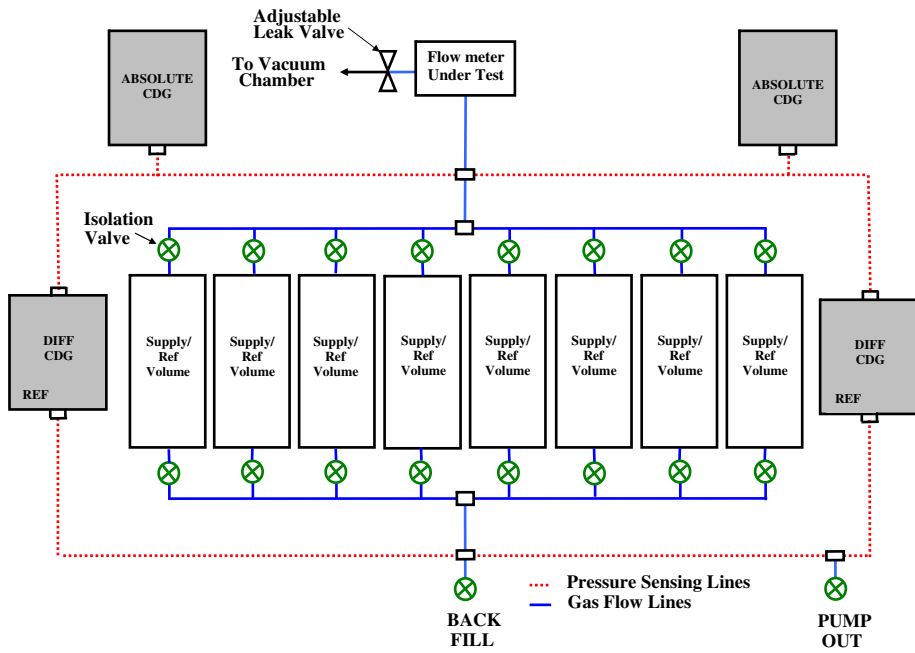


Fig. 1. Schematic of Precision Gas Flow Calibrator (PGFC).

Using seven of the cylinders to supply flow and one for pressure reference provides a maximum of 0.0265 m^3 of gas supply. Using two CDGs each for absolute and differential measurements provides for optimum resolution to be achieved by choosing an appropriate full scale range as well as the possibility of collecting redundant data to help fully characterize the measurement system. The supply pressure will ultimately be limited to 1333 kPa as this is the maximum line pressure allowable for high precision differential CDGs. The requirement determining measurement interval for the flow calibrator is that enough quality data be acquired in the time it takes the supply pressure to drop to 0.995 of its initial value (Eq. (1)).

Solving Eq. (1), assuming a measurement interval of 100 s for convenience, yields a valve conductance of $1.32 \times 10^{-6} \text{ m}^3/\text{s}$. A supply pressure of 133.3 kPa under these conditions will generate an average flow rate of $1.74 \times 10^{-6} \text{ m}^3/\text{s}$ with a total drop of 0.667 kPa. Raising the supply pressure to 1333 kPa therefore will generate roughly $1.74 \times 10^{-5} \text{ m}^3/\text{s}$ in flow for the same conditions with a 6.667 kPa total drop. These are maximum values for a 100 s interval. It is easily seen that there are many configuration options for addressing flow rates from 1.74×10^{-8} to $1.74 \times 10^{-5} \text{ m}^3/\text{s}$. Table 1 lists some possible configurations which can be used to generate known flow rates spanning three decades. A proposed design for a $1.74 \times 10^{-5} \text{ m}^3/\text{s}$ ($\sim 1000 \text{ sccm}$) PGFC and its basic operation are described below.

4. Proposed realization of a PGFC

The SCM is of all stainless steel construction utilizing metal gasketed connectors and valves. The eight stainless steel gas vessels are arranged in a symmetric array and

Table 1
Possible supply volume configurations and flow rates.^a

Supply volume (m^3)	Supply pressure (Pa)	$\Delta P/\Delta t$ (Pa/s)	Flow rate Q (m^3/s)
0.0038	10,000	0.50	1.87×10^{-8}
0.0076	13,330	0.667	4.98×10^{-8}
0.0151	13,330	0.667	9.96×10^{-8}
0.0038	133,300	6.667	2.49×10^{-7}
0.0076	133,300	6.665	4.98×10^{-7}
0.0151	133,300	6.665	9.96×10^{-7}
0.0038	10,00,000	50.00	1.88×10^{-6}
0.0076	13,33,000	66.65	4.98×10^{-6}
0.0114	13,33,000	66.65	7.47×10^{-6}
0.0265	13,33,000	66.65	1.74×10^{-5}

^a 100 s measurement interval.

mutually connected from a common junction through equal lengths of tubing. Four CDGs are also mutually connected through a junction just downstream of the vessel junction and symmetrically arranged within the vessel assembly. The symmetry and compactness of the design allows the entire assembly to be placed in a form fitting polystyrene foam enclosure roughly 0.75 m square by 1 m high. This ensures that variations in environmental temperature will have negligible effect during 100 s measurement intervals. Calibrated Platinum Resistance Thermometers (PRT) mounted on all cylinders and along the SCM will monitor temperature continuously to a resolution of better than $\pm 5 \text{ mK}$.

Output to the flow meter under test and leak valve is from a single port just downstream of the pressure sensing junction. The outlet of the leak valve will be connected to a pumping chamber whose pumping speed is sufficient for maintaining vacuum conditions for the flow rates generated.

Each of the seven possible configurations of supply volume, in multiples of 0.0038 m^3 , will need to be carefully calibrated. To calibrate a specific configuration, a calibrated reference vessel filled to a known pressure will be connected through an isolation valve to the output of the SCM. After the volume to be calibrated is evacuated, the isolation valve will be opened and the pressure allowed to equalize with the final pressure and temperature recorded. This will be repeated for a statistically significant range of pressures and the pressure ratios thus obtained will be used to determine the supply volume relative to the calibrated reference vessel. Once this is accomplished for each configuration it need only be repeated periodically based on the calibration cycle of the reference vessel as the dimensional stability of the components is very high.

To begin a flow meter calibration, the entire system is evacuated with the leak valve fully closed and the absolute CDGs zeroed. The system is then backfilled with the calibration gas to the pressure required to realize the low end of the flow range for the flow meter under test. The differential CDGs will then be zeroed and the flow calibrator partitioned into supply and reference volumes. The choice of supply volume configuration will depend on the range of calibration, response time and overall resolution of the flow meter under test. Once all thermometers are stabilized, the leak valve will be slowly opened to establish the desired flow rate. The peak pressure reached in the output vacuum chamber can be used to estimate flow rate. After the flow rate stabilizes, the reference volume pressure will be adjusted to establish a pressure differential commensurate with the 0.5% limit for the supply volume in use and then will remain connected only to the reference side of the differential CDGs. When the supply pressure has dropped by 0.5% data taking ends and the parameters for the next data point are set. A linear fit made to the differential pressure between the reference and supply volumes versus time determines the average rate of supply pressure drop. The readout of the flow meter under test is also recorded synchronously. The slope during the measurement interval multiplied by the value of the supply volume (including the volume of the flow meter under test) is equated to the average readout of the flow meter under test during the same measurement interval where:

$$Q = V \frac{dP}{dt} \quad (5)$$

The slope of the line fit $\frac{dP}{dt}$ is equivalent to the average of the individual measurements of $\frac{\Delta P}{\Delta t}$ over the measurement interval. Mass flow rate can be determined from the temperature and average pressure of the gas over the same interval. The entire range of calibration will be covered by progressing upwards in supply pressure and thus flow rate in a similar fashion.

The choice of CDG range will depend upon the maximum supply pressure and the maximum differential pressure. In a $1.74 \times 10^{-6} \text{ m}^3/\text{s}$ configuration with a 0.0265 m^3 supply volume for example, a 133.32 kPa full scale absolute CDG can be used along with a 1.33 kPa full scale differential CDG. In addition, 1333.2 kPa absolute and 13.33 kPa differential CDGs can also be used in parallel as four measurement ports are available. This would also allow for

continuous operation to $1.74 \times 10^{-5} \text{ m}^3/\text{s}$ by using isolation valves to protect the 1.33 and 133.3 kPa CDGs from overpressure as the supply pressure is increased (volume value adjusted accordingly).

Many combinations of supply volume and pressure, time interval and valve conductance can be used in generating a single range of flow rates. The system can therefore be optimized to adjust the calibration uncertainty to match the resolution of the flow meter under test and fully characterize its behavior.

5. Preliminary estimate of system uncertainty

The analysis of uncertainty follows the guidelines described in NIST Technical Note 1297 [4] which groups the components of measurement uncertainty into two categories: type “A” – those evaluated by statistical methods and type “B” – those evaluated by other means. The estimated system uncertainty derives primarily from the measurements used to determine the flow rate, Q , as defined in Eq. (5): where V is the calibrated supply volume and $\frac{dP}{dt}$ is obtained from the slope of the line fit to individual measurements of ΔP versus Δt .

Similar uncertainties as obtained in Ref. [1] for the supply volume calibration measurements are assumed for the PGFC. The calibration performed on the supply volume in Ref. [1] had a standard deviation of 0.21% and was evaluated by statistical means. It is thus treated as a type “A” uncertainty. The 1 l reference vessel used to calibrate the precision gas flow meter volume in Ref. [1] had a vendor calibration uncertainty of 0.08% which is treated as a type “B” as this is the maximum value (rectangular distribution).

The CDG manufacturer states a calibration uncertainty of 0.05%, however individual pressure drops as measured in Ref. [1] showed an overall reproducibility in ΔP of $\pm 0.25\%$ for similar ranges which can most probably be attributed to drift, resolution, and noise effects. This is taken as the uncertainty in ΔP due to the measuring instruments (which are identical to those in Ref. [1]). The measurement of ΔP ($P_0 - P$), assumes a constant reference volume pressure P_0 . However, the basic measurement uncertainty of $\pm 5 \text{ mK}$ in the temperature of the reference volume introduces an uncertainty in reference pressure of 0.016%; for an overall pressure drop of 0.5% within a measurement interval this propagates as an additional uncertainty in ΔP of 0.32% due to uncertainty in the measurement of reference volume temperature. Effective thermal isolation of the gas vessels from the environment and their close proximity within the thermal enclosure ensures any drift which does occur during a measurement interval is common to all cylinders with no overall effect on ΔP measurements. These uncertainties represent the maximum contribution throughout the calibration range and are treated as type “B” components. The response time of the CDG controller is typically set at 1 ms. The error for readings taken 1 s apart, assuming $\pm 0.5 \text{ ms}$ error for each measurement, is therefore 0.1% for Δt ; this is also treated as a type “B” uncertainty.

As the rate of pressure drop is basically exponential, as described above, and assuming parameters are adjusted

Table 2

Precision gas flow calibrator estimated uncertainties.

Source	Type	Combined as	Magnitude (%)
Flow calibrator volume	A ₁	A ₁	0.21
Calibration volume	B ₁	B ₁ /√3	0.05
ΔP	B ₂	B ₂ /√3	0.14
Δt	B ₃	B ₃ /√3	0.06
Non-linearity	B ₄	B ₄ /√3	0.14
Temperature	B ₅	B ₅ /√3	0.18
Total uncertainty U		√{A ² + Σ (B _i /√3) ² }	0.35
Expanded uncertainty	U _T = kU (k = 2)		0.70

such that the overall drop is limited to 0.5% or below during a measurement, the first non-linear term remains at 0.25% or less of the linear term. As this represents the maximum contribution throughout the calibration range it is treated as a type “B” component.

The outlet port of the gas flow calibrator will be connected to the flow meter under test using an estimated 50 cm of 1.27 cm diameter stainless steel tube. Using Poiseuille’s Law and assuming maximum flow rate and a supply pressure of 1333 kPa, the total pressure drop from the outlet port to the flow meter under test will be less than 0.133 Pa and thus negligible. All other paths within the SCM are no longer than 20 cm and of the same diameter thus exhibiting even smaller drops. The flow meter under test will introduce a pressure drop and add a small amount to the supply volume as well. The overall effect will be to lower the flow rate; the precision of the measurements should not change. The average pressure within

the flow meter under test will be used to calculate the mass flow rate. The individual uncertainties and the overall system uncertainty as calculated according to NIST Technical Note 1297 are shown in Table 2. The expanded uncertainty (which represents a confidence level of approximately 95%) is 0.70%. This does not include the contribution from reproducibility of the flow meter under test.

6. Conclusions

A gas flow calibrator capable of generating a maximum of $1.74 \times 10^{-05} \text{ m}^3/\text{s}$ has been proposed. Based upon a proven design and using commercially available components and instrumentation, it can be readily implemented in an industrial environment. It is estimated that flow rates comparable with small to medium piston provers and the minimum flows addressed by PVTt systems can be realized to within a $k = 2$ uncertainty of 0.70%. The reproducibility inherent to a well-defined exponential decay as well as the simplicity and ruggedness of the design should provide a stable, high precision standalone reference for gas flow calibration.

References

- [1] P.D. Levine, J.R. Sweda, A precision gas flowmeter for vacuum calibration, *J. Vac. Sci. Technol. A* 15 (3) (1997) 747–752.
- [2] J.D. Wright, G.E. Mattingly, NIST Calibration Services for Gas Flow Meters, NIST Special Publication 250-49, August 1998.
- [3] John D. Wright, Aaron N. Johnson, Michael R. Moldover, Gina M. Kline, Gas Flowmeter Calibrations with the 34 L and 677 L PVTt Standards, NIST Special Publication 250-63.
- [4] B.N. Taylor, C.E. Kuyatt, Guidelines for Evaluating and Expressing Uncertainty of NIST Measurement Results, NIST TN 1297.1994 edition.